2.1

$$y = [1 \;\; x \;\; x^2 \;\; \cdots \;\; x^M] \begin{bmatrix} \beta_0 \\ \vdots \\ \beta^M \end{bmatrix}$$

Root Mean Squared Error

$$E_{RMS} = \sqrt{\frac{f(\beta)}{n}}$$

$M \Uparrow \longrightarrow$ overfitting

Regularization

$$J[f] = f(\beta) + \lambda \sum_j |\beta_j|^q$$

$$= \|y - X\beta\|^2 + \lambda \|\beta\|^2$$

$$= (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta$$

$$\nabla_\beta J[f] = \nabla_\beta \left(\cancel{y^T y} - \underbrace{y^T X \beta}_{-X^T y} - \underbrace{\beta^T X^T y}_{-X^T y} + \underbrace{\beta^T X^T X \beta}_{2X^T X \beta} \right) + \nabla_\beta \underbrace{\lambda \beta^T \beta}_{2\lambda\beta}.$$

$$= 0 \implies \left[X^T (Y - X\beta) = \lambda \beta \right]$$

$x^T\beta \longrightarrow \phi(x)^T\beta$

low dimension

high dimension

<kernel>

$$k(x_1, x_2) = \phi(x_1)^T \phi(x_2)$$

"inner-product"

- $J[f] = \|y - \phi(X)^T\beta\|^2 + \lambda\|\beta\|^2$

$$= y^Ty - y^T\phi(X)^T\beta - \beta^T\phi(X)y + \beta^T\phi(X)\phi(X)^T\beta + \lambda\beta^T\beta$$

- $\nabla_\beta J[f] = 0 - \phi(X)y - \phi(X)y + 2\phi(X)\phi(X)^T\beta + 2\lambda\beta$

$$= -2\phi(X)y + 2\phi(X)\phi(X)^T\beta + 2\lambda\beta$$

$$= 0 \quad (\text{극값})$$

- $\beta^* = (\lambda I + \phi(X)\phi(X)^T)^{-1}\phi(X)y$

$$= \phi(X)(\lambda I + \phi(X)^T\phi(X))^{-1}y$$

$$= \phi(X)\underbrace{\boxed{(\lambda I + K(X,X))^{-1}y}}_{\alpha^*}$$

- $f^*(x) = \phi(x)^T\beta^*$

$$= K(x, X)\cdot\alpha^*$$

example let $k(x,z) = (x^T z)^2$

$$= \langle \underbrace{(x_1^2,\ x_2^2,\ \sqrt{2}\,x_1 x_2)}_{\phi(x)},\ \underbrace{(z_1^2,\ z_2^2,\ \sqrt{2}\,z_1 z_2)}_{\phi(z)} \rangle$$

let $X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $X_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ $Y = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

then $\beta^* = \phi(X) \cdot \alpha^*$

$$= \phi(X)\,(\lambda I + k(X,X))^{-1}\, y.$$

$$k(X,X) = \begin{bmatrix} (1\cdot1+1\cdot1)^2 & (2\cdot1+0\cdot1)^2 \\ (1\cdot2+1\cdot0)^2 & (2\cdot2+0\cdot0)^2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 16 \end{bmatrix}$$

$$\lambda I + k(X,X) = \begin{bmatrix} 5 & 4 \\ 4 & 17 \end{bmatrix}, \quad \alpha^* = \begin{bmatrix} 5 & 4 \\ 4 & 17 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \frac{1}{69} \cdot \begin{bmatrix} 43 \\ -2 \end{bmatrix}$$

$(\lambda = 1)$

$$f^*(x) = k(x,X) \cdot \alpha^*$$

$$= [1, 4] \cdot \frac{1}{69} \begin{bmatrix} 43 \\ -2 \end{bmatrix}$$

$x^T = [1, 0]$

cf. $k(x,z) = \exp\left(-\frac{\|x-z\|^2}{2\sigma^2}\right)$

$$k_2(x,z) = \frac{k_1(x,z)}{\sqrt{k_1(x,x)}\sqrt{k_1(z,z)}}$$

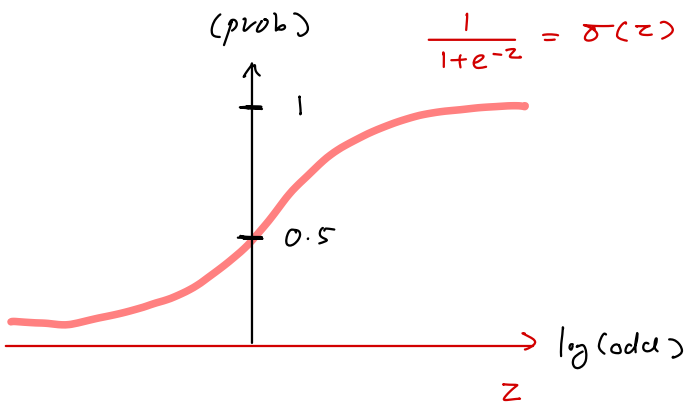
(sigmoid)

$$\begin{cases} \text{probability} & p = \dfrac{e^z}{1+e^z} \in [0, 1] \\ \text{odds} & o = e^z \in [0, \infty) \end{cases}$$

$$o = \frac{p}{1-p} \qquad p = \frac{o}{1+o}$$

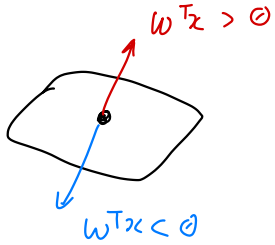
$\downarrow$

log (odds) z



Logistic Regression (MLE)

. hyperplane $w^T x = 0$



. 정답. $y_i \in \{0, 1\}$

예측 $\sigma(w^T x) \in [0, 1]$

log (odd)

. $P(y_i = 1 \mid x_i, w) = \sigma(w^T x_i)$

. $\mathcal{L} = \prod_i P(y_i \mid x_i, w)$

$$= \prod_i P(y_i = 1 \mid x_i, w)^{y_i} \, P(y_i = 0 \mid x_i, w)^{1-y_i}$$

$$-\log \mathcal{L} = -\left[\sum_i y_i \log \frac{1}{1+e^{-w^T x_i}} + \sum_i (1-y_i) \log \frac{e^{-w^T x_i}}{1+e^{-w^T x_i}} \right]$$

$$= \sum_i \log(1+e^{-w^T x_i}) + \sum_i (1-y_i) w^T x_i$$

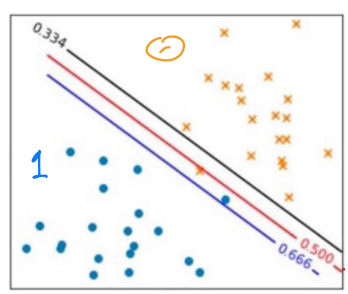
$$\nabla_w(-\log \mathcal{L}) = \sum_i \frac{e^{-w^T x_i}}{1+e^{-w^T x_i}} \cdot (-x_i) + \sum_i (1-y_i) x_i$$

$$= \sum_i \left[\sigma(w^T x_i) - y_i \right] x_i$$

예측 확률

실제

방향



빨간 선 $w^T x \geq 0$ prob $\geq 50\%$

파란 선 $w^T x \geq ?$ prob $\geq 66\%$

claim $J = -\log \mathcal{L}$ 일 때, J는 convex하고 global min을 가진다.

$$J = -\log \mathcal{L} = \sum l_i$$

$$\nabla_w l_i = \left(\sigma(w^T x_i) - y_i\right) x_i$$

$$\nabla_w J = \sum_i \left(\sigma(w^T x_i) - y_i\right) x_i$$

Note. 하나에서 성립하면 대체로 성립한다.

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

- $\nabla_w(\nabla_w l_i) = \left[\nabla_w\left(\sigma(w^T x_i) - y_i\right)\right] \cdot x_i^T$

$$= \sigma'(w^T x_i) \cdot x_i x_i^T$$

Note. $\sigma' = \sigma(1-\sigma) \geq 0$

- $\nabla_w(\nabla_w J) = \sum_i \sigma'_i \, x_i x_i^T$

$$= \underbrace{\begin{bmatrix} | & & \\ x_1 & \cdots & \\ | & & \end{bmatrix}}_{X^T} \underbrace{\begin{bmatrix} \sigma' & & \\ & \ddots & \\ & & \sigma' \end{bmatrix}}_{S} \underbrace{\begin{bmatrix} - & x_1 & - \\ & \ddots & \end{bmatrix}}_{X}$$

- for $\forall v \in \mathbb{R}^d$

$$v^T(\nabla_w^2 J)v = v^T X^T S X v$$

$$= (Xv)^T S (Xv)$$

$$= \sum_i \underbrace{(\sigma'_i)}_{\geq 0} \underbrace{\left((Xv)_i\right)^2}_{\geq 0} \geq 0$$

positive semi-definite!

$\langle v, Hv \rangle \geq 0$

$\Longleftrightarrow$ J convex : global min

$$\boxed{\text{multiclass classification}: \quad p(y=k \mid x, W) = \frac{e^{w_k^T x}}{\sum_j e^{w_j^T x}}}$$

assume $x = \begin{bmatrix} 5.1 \\ 3.5 \\ 1.4 \\ 0.2 \end{bmatrix}$ $w_1 = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.3 \\ 0.4 \end{bmatrix}$ $w_2 = \begin{bmatrix} -0.3 \\ 0.1 \\ 0.5 \\ 0.2 \end{bmatrix}$ $w_3 = \begin{bmatrix} 0.2 \\ -0.4 \\ 0.1 \\ 0.3 \end{bmatrix}$

then
$$\begin{cases} z_1 = w_1^T x = 1.71 \\ z_2 = w_2^T x = -0.44 \\ z_3 = w_3^T x = -0.18 \end{cases} \qquad \begin{cases} p_1 = 0.789 \\ p_2 = 0.091 \\ p_3 = 0.12 \end{cases}$$

let $y = [1, 0, 0]$ then $p(y \mid x, W) = p_1^{1} \, p_2^{0} \, p_3^{0}$

$= 0.789$

multiclass classification.

- $$\boxed{p_{ik} = P(y_i = k \mid x_i, W) = \frac{e^{w_k^T x_i}}{\sum_j e^{w_j^T x_i}}}$$

$$W = \begin{bmatrix} - & w_1^T & - \\ & \ddots & \\ & & \end{bmatrix}$$

- $\mathcal{L} = P(y_1, \dots, y_n \mid x_1, \dots, x_n, W)$

$$= \prod_{i=1}^{N} p(y_i \mid x_i, W)$$

$$= \prod_{i=1}^{N} \prod_{k=1}^{C} \underbrace{p(y_i = k \mid x_i, W)}_{=: p_{ik}}^{y_{ik}}$$

$$y_{ik} = \begin{cases} 1 & y_i = k \\ 0 & y_i \neq k \end{cases}$$

- $$-\log \mathcal{L} = -\sum_{i=1}^{N} \sum_{k=1}^{C} y_{ik} \log p_{ik}$$

- $$p_{k} = \frac{e^{w_k^T x}}{\sum e^{w_j^T x}} = \frac{e^{z_k}}{\sum e^{z_j}}$$

class

- $$\frac{\partial}{\partial w_{mn}} p_{k} = \frac{\underbrace{\boxed{\frac{\partial e^{z_k}}{\partial w_{mn}}}}_{= \delta_{mk} \cdot e^{z_k} \cdot x_n} \cdot \left(\sum e^{z_j}\right) + \left(e^{z_k}\right) \cdot \underbrace{\boxed{\frac{\partial \sum e^{z_j}}{\partial w_{mn}}}}_{= e^{z_m} \cdot x_n}}{\left(\sum e^{z_j}\right)^2}$$

class

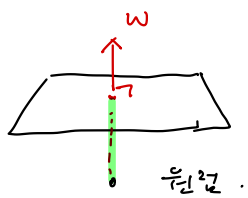
$$= p_k \left(\delta_{mk} - p_m\right) \cdot x_n$$

- $$\frac{\partial}{\partial w_{mn}} (-\log \mathcal{L}) = -\sum_{i}^{N} \sum_{k}^{C} y_{ik} \cdot \frac{1}{\cancel{p_{ik}}} \cdot \cancel{p_{ik}} \left(\delta_{mk} - p_{im}\right) x_{in}$$

$$= \sum_{i}^{N} \left(p_{im} - y_{im}\right) x_{in}$$

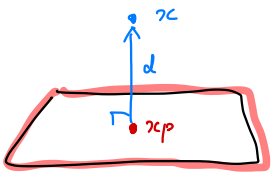
예측 정답

2.2



$$w^T x + b \begin{cases} > 0 & y = 1 \quad \text{Positive} \\ < 0 & y = -1 \quad \text{Negative} \end{cases}$$

hence $y(w^T x + b) \geq 0$ 중요



hyperplane

$$0 = w^T x_p + b$$
$$= w^T (x - d) + b$$
$$= w^T (x - \alpha w) + b$$

- $\|d\| = \alpha \cdot \|w\|$
  $= (w^T x + b) / \|w\|$

- margin

  : $r(w, b) = \min_{x \in D} \dfrac{|w^T x + b|}{\|w\|}$

"margin을 최대화하는 것이 SVM의 목표이다."

$$\underset{w,b}{\operatorname{argmax}} \left[\min_{x} \frac{\overset{1.}{\cancel{|w^T x + b|}}}{\|w\|} \right] = \underset{w,b}{\operatorname{argmin}} \|w\|$$

$$\boxed{\begin{aligned} \text{s.t} \quad & ① \ \forall i \quad y_i (w^T x_i + b) \geq 0 \\ & ② \ \min_{x} |w^T x + b| = 1 \\ \Longleftrightarrow \ & \forall i \quad y_i (w^T x_i + b) \geq 1 \end{aligned}}$$

$$\text{minimize} \quad \frac{1}{2}\|w\|^2 + \max_{\alpha_i \geq 0} \alpha_i \left(1 - y_i (w^T x_i + b)\right)$$

$$\max_{\alpha \geq 0} \alpha \left(1 - y(w^T x + b)\right) = \begin{cases} 0 & y(w^T x + b) \geq 1 \\ \infty & \text{else.} \end{cases}$$

- $$\min_{w,b} \max_{a_i \geq 0} \left[\frac{1}{2}\|w\|^2 + \sum_i \alpha_i \left(1 - y_i(w^T x_i + b)\right) \right]$$

$$\geq \max_{a_i \geq 0} \min_{w,b} \left[\frac{1}{2}\|w\|^2 + \sum_i \alpha_i \left(1 - y_i(w^T x_i + b)\right) \right]$$

- $$= \frac{1}{2} w^T w + \sum_i \alpha_i - w^T \left(\sum_i \alpha_i y_i x_i \right) - \sum_i \alpha_i y_i b$$

$$\begin{cases} \nabla_w = 0 & w = \sum_i \alpha_i y_i x_i \\ \dfrac{\partial}{\partial b} = 0 & \sum_i \alpha_i y_i = 0 \end{cases}$$

- $$\boxed{\max_{\alpha_i} \sum_i \alpha_i - \frac{1}{2} \overbrace{\sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i^T x_j}^{w^T w}} \quad \text{s.t.} \quad \begin{array}{l} ① \ \alpha_i \geq 0 \\ ② \ \sum \alpha_i y_i = 0 \end{array}$$

- support vectors I.

① $\alpha_i > 0$ then $i \in I$

② obtain $w = \sum_{i \in I} \alpha_i y_i x_i$

③ obtain $b = y_i - x_i^T w \quad (i \in I)$

④ $$\|w\|^2 = \overbrace{\left[\sum_{i \in I} \alpha_i y_i x_i^T \right]}^{w^T} w \quad 1 = y_i(x_i^T w + b)$$

$$= \sum_{i \in I} \alpha_i (1 - y_i b)$$

$$= \sum_{i \in I} \alpha_i - b \cancel{\sum_{i \in I} \alpha_i y_i}$$

$$= \sum_{i \in I} \alpha_i$$

⑤ $$\text{margin} = \frac{|x^T w + b|}{\|w\|} = \frac{1}{\sqrt{\sum_{i \in I} \alpha_i}}$$